Douglas Liao

+46 727333218 | douglasliao77@gmail.com | linkedin.com/in/douglasliao | https://github.com/douglasliao77

OVERVIEW

MSc candidate in Computer Science and Engineering, with 2 years of experience at Ericsson. Strong foundation in the full software development life-cycle, including design, implementation, and testing of web applications. Driven by a desire to develop efficient, robust applications that solve real-world problems.

EDUCATION

Linköping University **Linköping, Sweden**

MSc in Computer Science and Engineering

Master's thesis, Ericsson

- Conducted end-to-end research on anomaly detection in Radio Access Networks using unsupervised AI models, achieving up to 83% F1-score in detecting network contention.

Bachelor's thesis, Personalkollen

- Architected and designed a log system in a cross-functional team, leading the technical decisions and aligning implementation with current system and requirement improving monitoring capabilities.

PROFESSIONAL EXPERIENCE

Ericsson **Linköping, Sweden**

Software Developer Intern, Cloud RAN ***June 2023–January 2025***

- Created a pipeline utilizing Docker to automate Pytest for testing upon submitting to the master branch in Gerrit, improving code quality with approximately 20%.
- Designed and implemented a real-time web application that transformed data into clear product performance insights, helping managers reduce reporting time by over 30%.
- Built an interactive visualization platform for 4G/5G base station simulations, reducing analysis time by over 35%.

Linköping University **Linköping, Sweden**

Course Assistant, Department of Computer science ***August 2022–January 2023***

- Led weekly seminars for 25+ students, contributed to a collaborative learning environment in a course that achieved an average feedback score over 4.0 out of 5.
- Conducted 10 hours of lab demonstrations weekly and participated in grading meetings, helping to standardize assessments and improve lab instruction quality.

Research Assistant, Department of Computer science ***June 2022–January 2023***

- Collaborated on AI/ML research projects, to reproduce and evaluate models from academic papers, validating their accuracy and applicability for further development.

PROJECTS

Optimized Electric Motor Usage

- Developed a dynamic programming solution to cut environmental costs by over 20% in a hybrid vehicle through optimal mode selection across road segments based on battery level and terrain.

Smart Biofuel Monitoring

- Implemented a heuristic algorithm showcasing possible camera placement strategies for biofuel monitoring systems, identifying cost-effective solutions in the 70-90% area coverage.

SKILLS & CERTIFICATION

Technical Skills: C++, Python, C, Java, HTML, CSS, JavaScript, VHDL, R Programming, SQL, React.

Certificate: Google Data Analysis, BCG Strategy Consulting Simulation (Forage).